JPX Investor Day 2024

Clearing Services

Japan Securities Clearing Corporation
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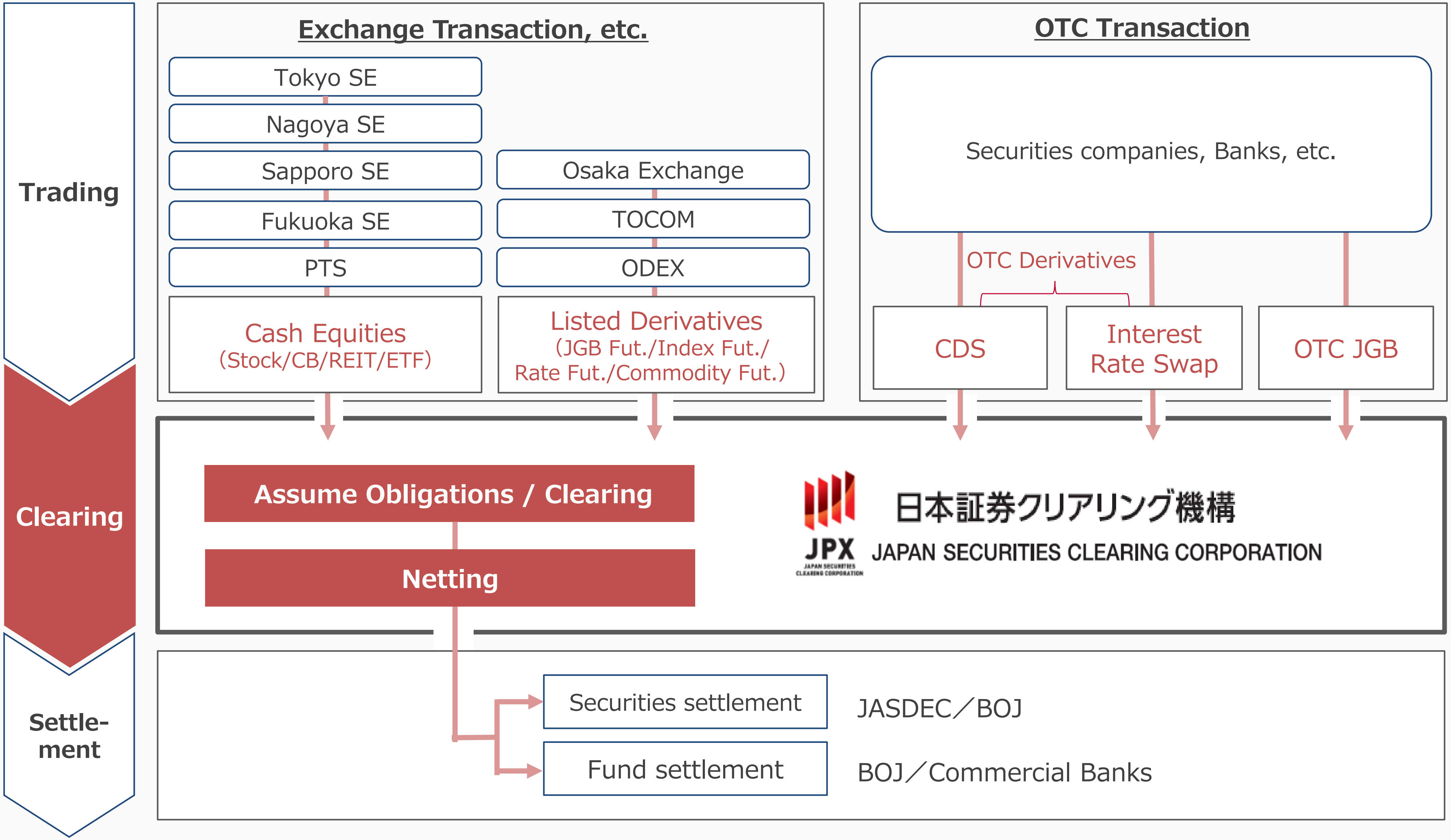
JSCC Overview

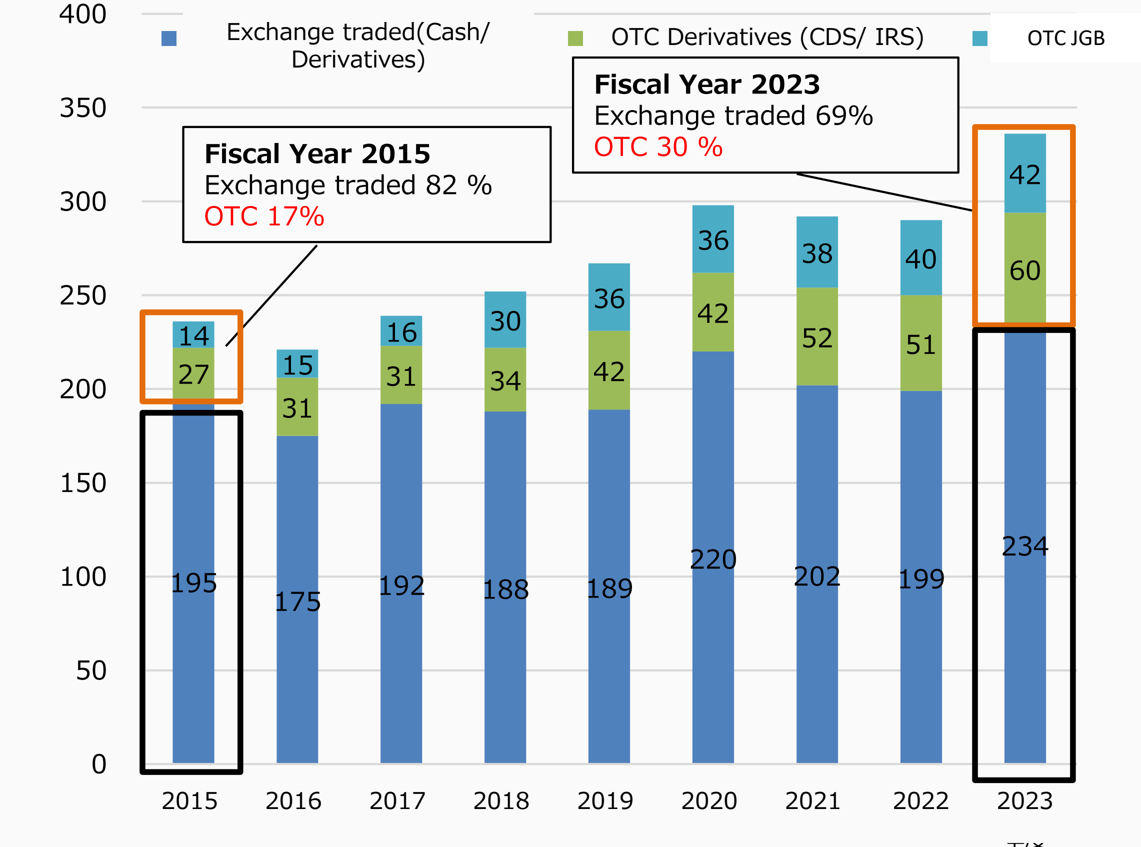
- **JSCC offers clearing functionality for a wide variety of transactions in Japanese domestic financial and commodity markets.**



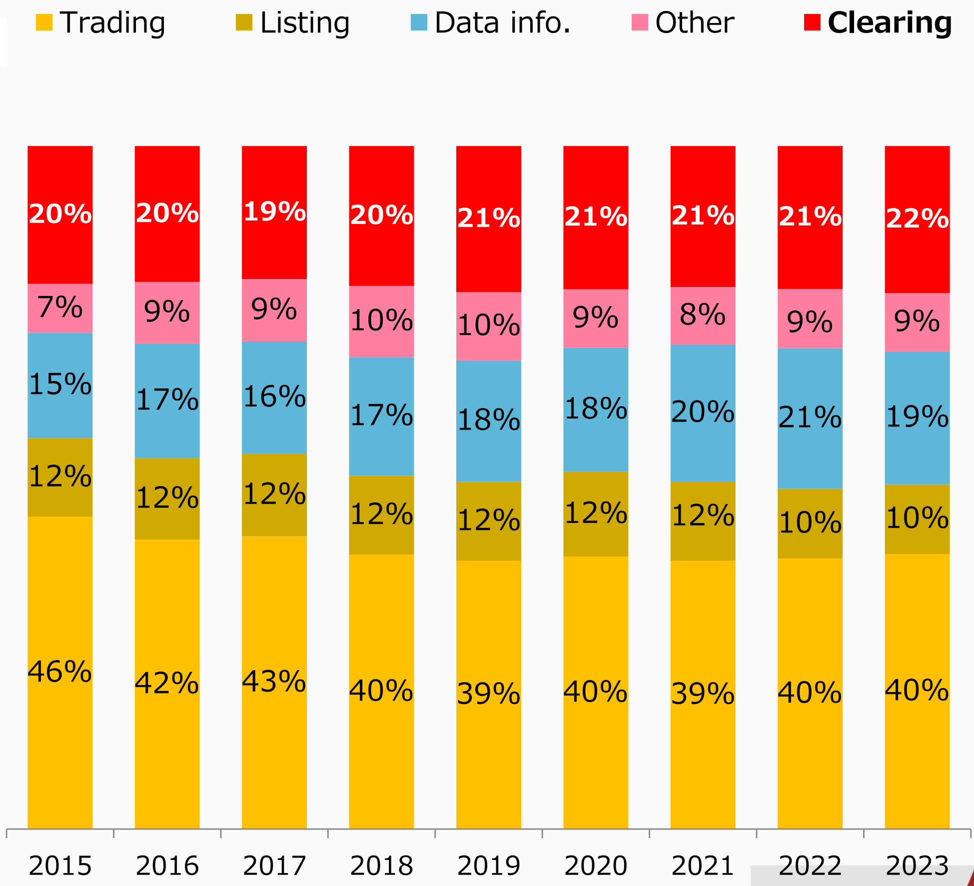
Revenue

- **<u>Operating revenues are expanding moderately,</u> mainly from improved market conditions and increase in OTC derivatives clearing volume.**
- **Majority of the revenues are from exchange-traded transactions (equities and listed derivatives), but <u>revenue from off-exchange transactions (OTC derivatives and OTC JGBs) has increased in recent years.</u>**
- **<u>JSCC's operating revenue (i.e., clearing-related revenue) accounts for approx. 20% of JPX's overall operating revenue.</u>**

<u>Breakdown of JSCC's operating revenue</u>

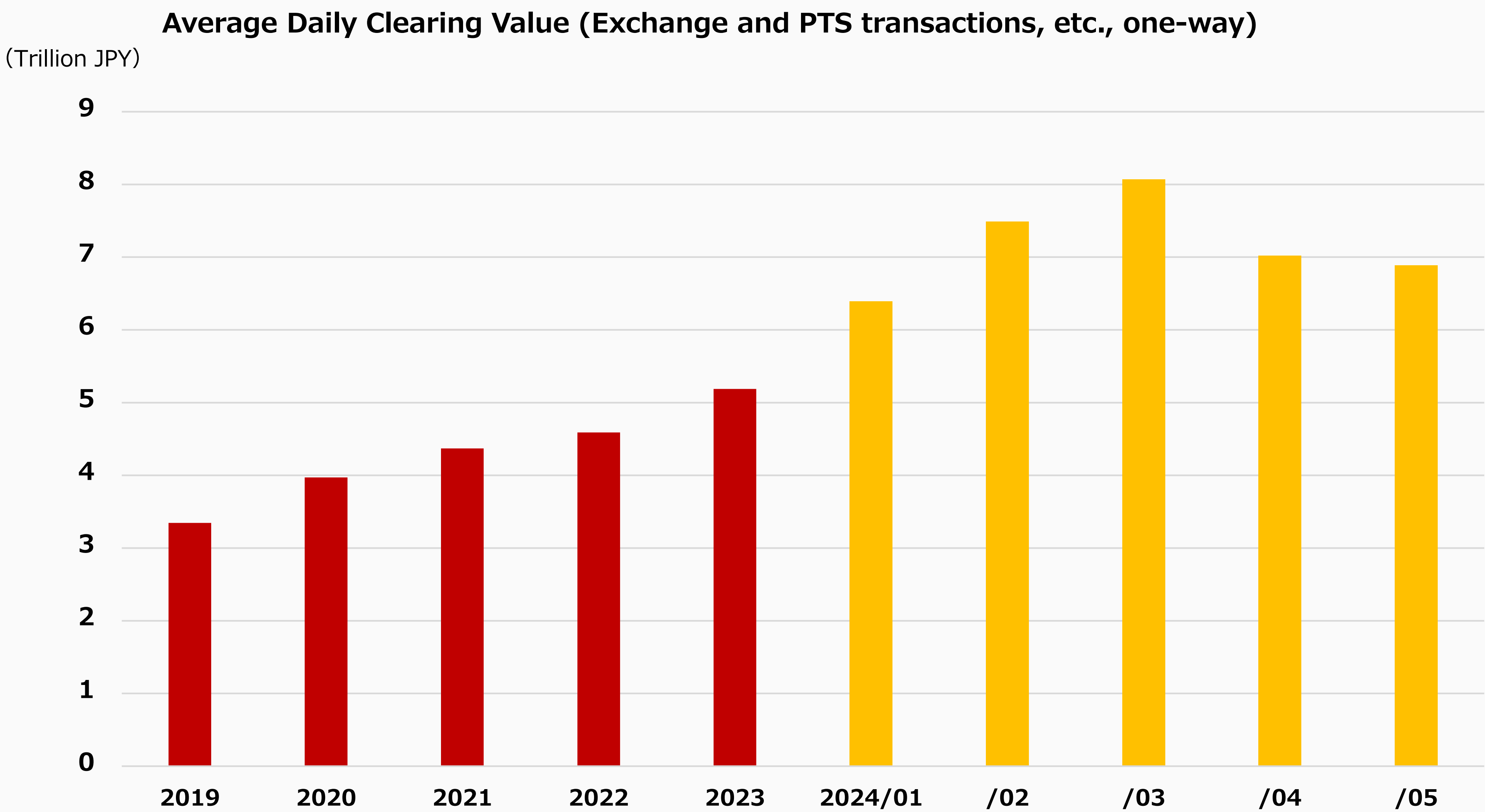


<u>Clearing-related revenue to JPX's overall operating revenues</u>



(Reference) Cash Equities Clearing Value

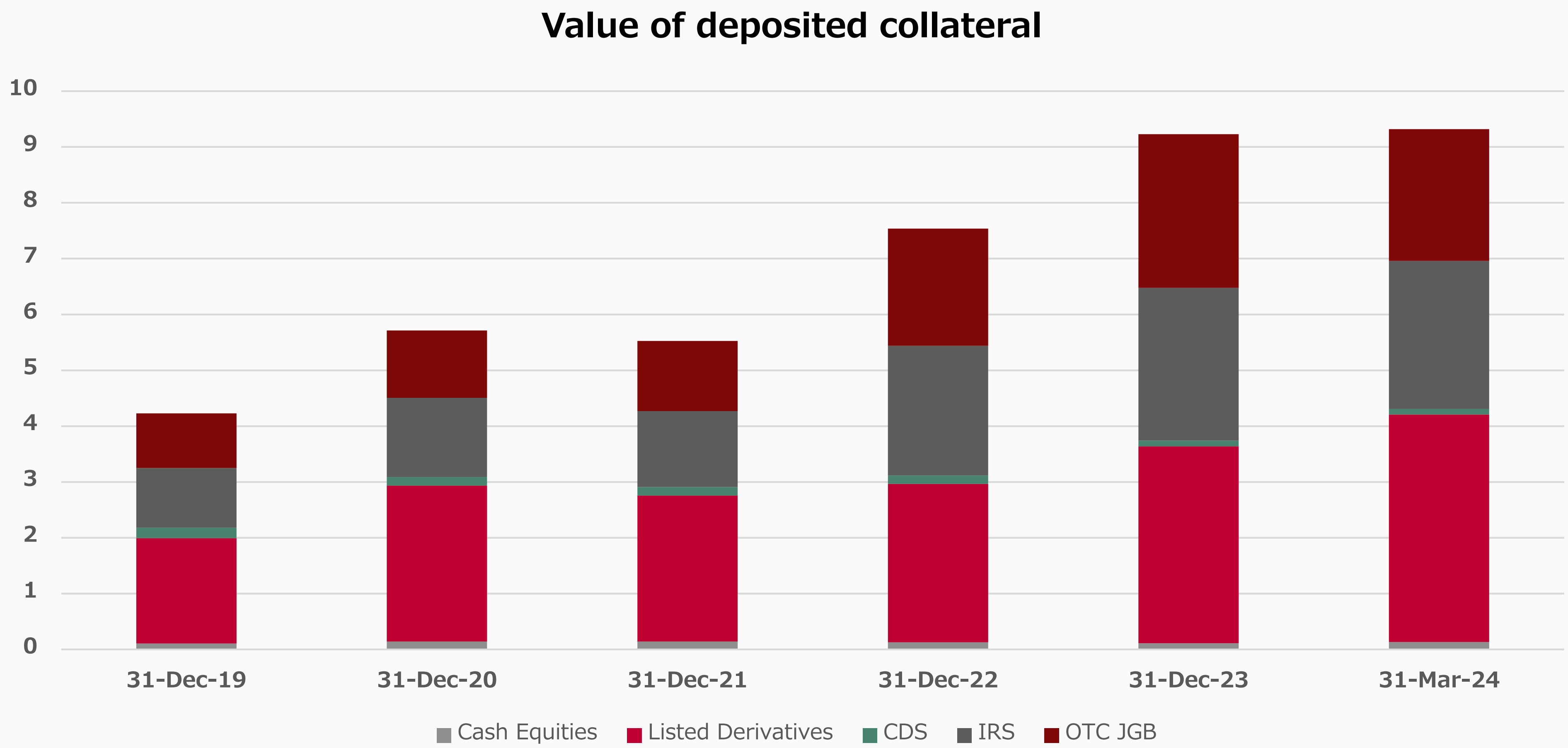
- **The clearing value has been steadily increasing over the past few years, with a particularly remarkable increase since the middle of last year**



(Reference)　Value of Deposited Collateral

- The value of collaterals deposited with JSCC has steadily increased with an expansion of the scope of clearing coverage and the clearing volume increase.
- (JSCC is currently studying how we can use deposited collaterals effectively to be ready for possible future rise in the level of short-term interest rates.)

(Trillion JPY)



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Competitive and Regulatory Environment

- **Competitive environment differs depending on cleared market.**

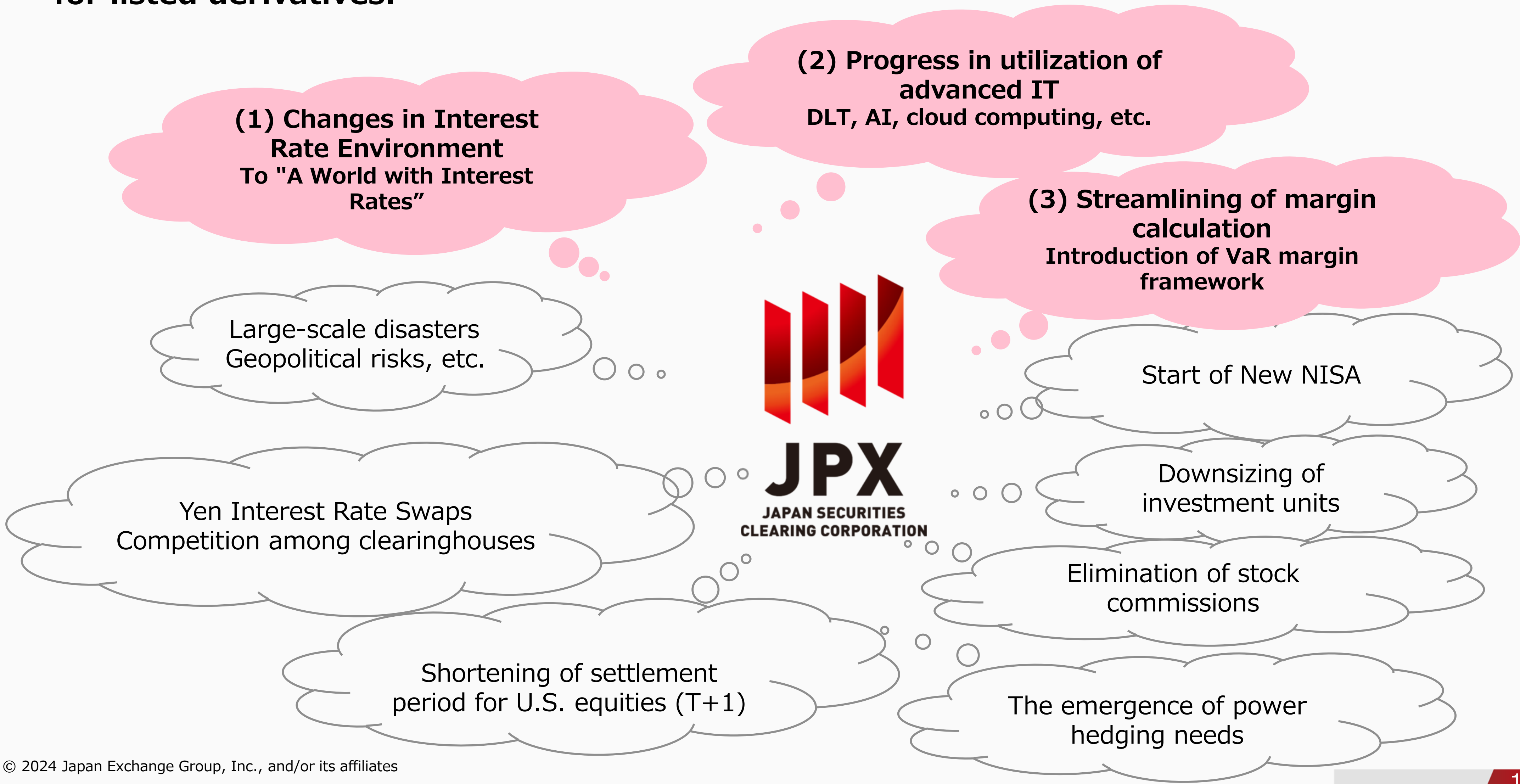
Market	Competitive Environment
Listed Cash Equities (Stock / ETF)	JSCC clears all trades on Japanese domestic exchanges and PTS
Listed Derivatives (Financial / Commodity)	**Exchanges/Clearinghouses compete globally** (CME/SGX)
OTC Derivatives (IRS / CDS)	**Intense global competition among clearinghouses** (LCH)
OTC JGB	JSCC clears most of the trades between domestic financial institutions

- **Regulatory environment: JSCC offers clearing functionality globally under supervision of the domestic and international/overseas regulators.**

Type of regulation	Detail
Global Regulation	**PFMI** (Principles for Financial Market Infrastructures) - International standards requiring higher standards for various items, such as risk management, with the aim to increase the robustness of FMIs - Compliance with the Principles is the minimum requirement for clearing transactions for global financial institutions.
Domestic regulations	Obtained licenses/permits under the **FIEA** and **CDA**
Foreign regulations	**Obtained CCP licenses from foreign authorities** for cross-border trade clearing (especially OTC derivatives) involving overseas financial institutions

Business Environment

- The business environment surrounding JSCC is ceaselessly changing, including JPY interest rate trend and progress in the use of distributed ledger technology (DLT).
- Today's presentation focuses on three topics: (1) changes in JPY interest rate environment, (2) progress in the use of advanced IT, and (3) the introduction of VaR margin framework for listed derivatives.

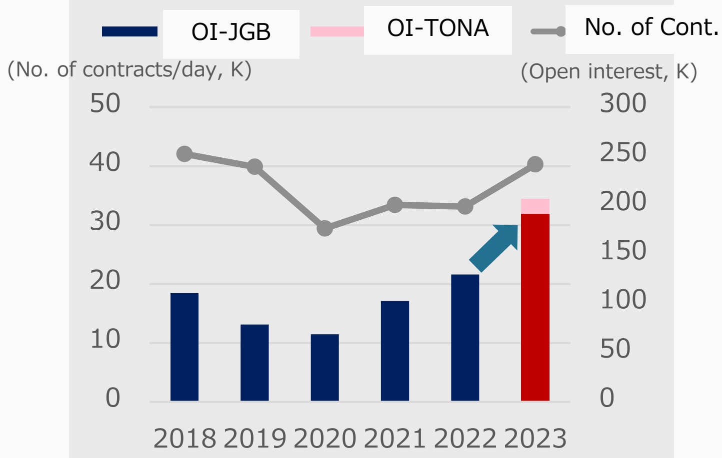


Changes in Interest Rate Environment
- Further Expansion of Clearing of JPY Interest Rate-related Products

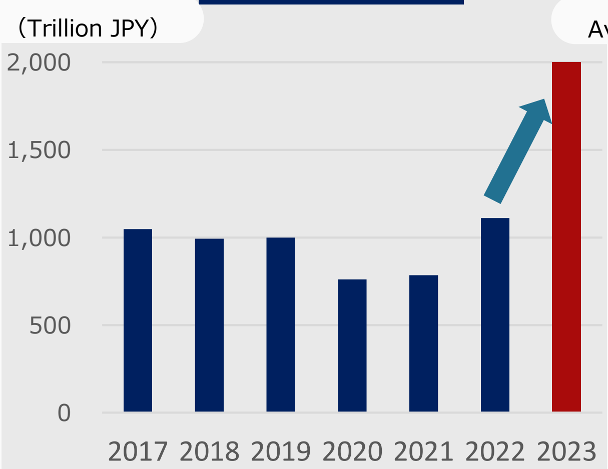
- Increased expectations of monetary policy revisions have attracted domestic and international attention, leading to **active trading and clearing of JPY interest rate-related products**
- Leveraging our strength as a one-stop provider of clearing services for JPY interest rate-related products, we will promote the following measures to revitalize the market
  1. **Cross-product sales activities** of JPY interest rate-related products **in cooperation with OSE**
  2. **Maximize margin efficiency** among JPY interest rate products (**expand the scope of cross margin** coverage to include **3-month TONA futures**)

1. Cross-product sales activities of JPY interest rate-related products

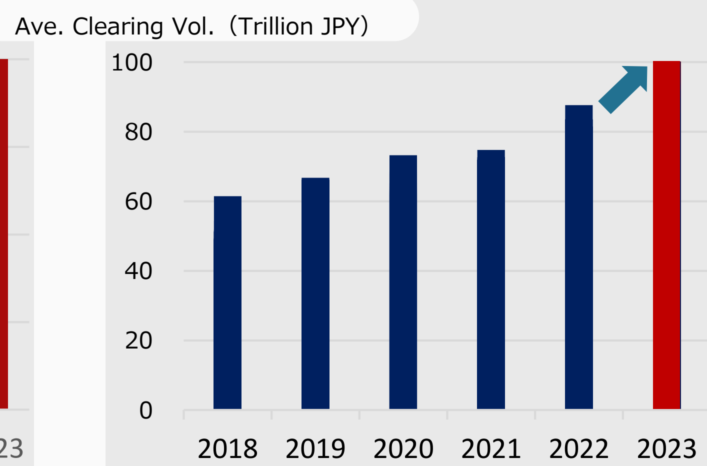
JGB and ST Rate Fut. Clearing



IRS Clearing



OTC JBG Clearing



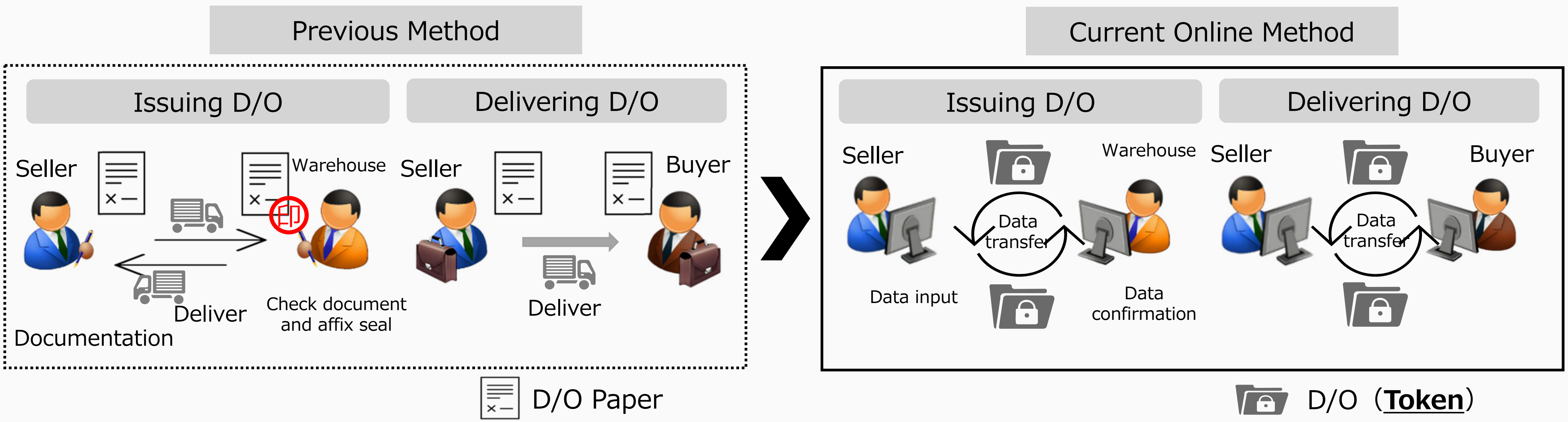
2. Margin efficiency among JPY interest rate products

Expansion of cross margin coverage	
IRS－ JGB Fut.	Already covered
IRS－TONA 3M Fut.	**Added from Mar. 2024**

Progress in Use of Advanced IT
- DLT Utilization in Delivery/Settlement of Commodity Futures

- JSCC **pursues the use of new technologies in existing operations**, such as settlement by delivery, from the perspective of improving operational efficiency.
- In January 2023, **tokenization of delivery order for rubber futures' settlement by delivery was realized, which is the first case of practical implementation of DLT.**
- Digitalized settlement by delivery with warehouse receipts in precious metal futures trading is also under consideration.
  - Since warehouse receipts are categorized as securities, unlike delivery orders, the legal system needs to be revised to allow for their digitalization.
- While keeping a close eye on the discussions on the revision of the legal system, we are **preparing for the realization of digitalized delivery/settlement of precious metal futures**.

Delivery and settlement of rubber futures (image)



Introduction of VaR Margin Framework for Listed Derivatives

- In November 2023, the margin calculation method for listed derivatives transactions was shifted from SPAN® to VaR.

  (※) EUREX (Europe) and OCC (US) have already transitioned, and ICE (US) and CME are in the process of the phase-in transition.

  - Use of numerous historical data **to prevent rapid margin spikes and increase transparency**
  - **Advanced risk management** through portfolio-by-portfolio risk measurement (e.g., margin optimization)

(Reference) Examples of changes in margin levels and characteristics due to introduction of VaR method

#	Item	New method	Margin level changes/trends
1	When one futures contract is held alone.	Calculate margins separately for short and long	Margin increased for Nikkei 225 Futures from SPAN®, Decreased for 10Y JGB Futures (Short)
2	Margin by Contract Month	Refined calculation by contract month	Margin decreased for index futures and OPs in the forward contract months, Margin increased for 3-month TONA futures in the forward contract months
3	Margin for combined positions	Refining risk offsets	Marked decrease in margins for the combination of Nikkei 225 Futures and TSE Growth Market 250 Futures
		New risk offsetting combinations	Margin for TSE REIT Index Futures and Nikkei VI Futures decreased markedly when combined with Nikkei 225 Futures

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